SOFTWARE REQUIREMENTS SPECIFICATION

University Library Reservation and Borrowing System

Codeflow PSUT · Industry Practices Using the SDLC · Session: UI/UX Design · November 4, 2026

Version / status	0.9 — Draft for design phase (teaching case)	Scope	Catalogue, reservations, loans, notifications, librarian operations
Audience	Students, designers, developers, testers	Out of scope	Source code, SQL, hosting, e-books, inter-library loans, payments

How to read this document. Each requirement has a unique ID (e.g. `FR-09`) so designers, developers and testers can refer to it precisely. "Shall" means mandatory once approved. Policies marked Proposed are example values for teaching — they are **not** confirmed PSUT rules.

1 Project overview

System name: University Library Reservation and Borrowing System.

Description. Students can browse books, search the catalogue, check availability, reserve books and monitor their borrowing status through a digital system. **The student must physically visit the university library to collect and return books.** Librarians handle physical preparation, collection and return confirmation.

Objectives

- Reduce unnecessary library visits.
- Improve book discovery.
- Allow digital reservations.
- Prevent conflicting reservations.
- Improve loan tracking.
- Automate reminders and availability notifications.
- Improve librarians' operational efficiency.

Key terms

- **Book (title)** — a bibliographic record, e.g. "Clean Code".
- **Copy** — one physical item of that book with its own barcode.
- **Reservation** — a request that holds a specific copy for a student.
- **Loan** — a copy that has been physically handed over to a student.
- **Waitlist** — a queue of students waiting for any copy of a title.

2 User roles and permissions

Guest — anyone, not signed in. Can browse and search the public catalogue without authentication.

Student — an authenticated, enrolled student. Can reserve books, view loans, receive notifications and monitor fines.

Librarian — library staff. Views student reservations, prepares books, confirms collection and returns, manages copy availability and monitors borrowing periods.

Administrator — manages policies (loan length, fine rates), staff permissions and system configuration.

Capability	Guest	Student	Librarian	Admin
Browse & search the public catalogue	✓	✓	✓	✓
Reserve a copy / join a waitlist / cancel own reservation	—	✓ own	—	—

Capability	Guest	Student	Librarian	Admin
View loans, due dates, fines and notifications	—	✓ **own**	✓ **all students**	— (audit only)
Mark ready, confirm hand-over, record returns, manage copies	—	—	✓	—
Configure policies, fine rates, staff roles, calendars	—	—	—	✓

> **Why separate permissions?** The principle of *least privilege*: each role gets only what it needs. A student sees only their own data; a librarian cannot change the fine policy; an administrator does not process day-to-day loans.

3 Functional requirements

Functional requirements describe **what the system must do**. Each is written so a tester can verify it with a clear pass/fail result.

ID	Requirement	Role
Catalogue and search		
FR-01	Guests and authenticated students shall be able to browse the library catalogue.	Guest, Student
FR-02	Users shall be able to search by title, author, ISBN, category and keyword.	All
FR-03	The system shall display book details, including title, author, category and copy availability.	All
FR-04	The system shall distinguish available, reserved, borrowed and unavailable copies.	All
Authentication		
FR-05	Students shall authenticate through an approved university identity mechanism using their university student identity.	Student
FR-06	Only verified, active, eligible students shall be allowed to reserve books.	Student
FR-07	Guests shall not be required to authenticate to browse or search the public catalogue.	Guest
FR-08	The system shall preserve the selected book and return the student to their intended reservation action after successful authentication.	Student
Reservations		
FR-09	Eligible students shall be able to reserve available physical book copies.	Student
FR-10	The system shall prevent multiple active reservations for the same physical copy.	System
FR-11	The system shall support waitlisting when all eligible copies are unavailable.	Student
FR-12	Students shall be able to view, track and cancel their pending reservations, subject to university policy.	Student
FR-13	The system shall provide a confirmation containing reservation status and collection instructions.	System
Notifications		
FR-14	Students shall be notified when a reserved book is ready for physical collection.	System
FR-15	Students on a waitlist shall be notified when a copy becomes available for reservation or allocation.	System
FR-16	Students shall receive reminders three days, two days and one day before the due date, and on the due date.	System
FR-17	Students with overdue loans shall receive daily reminders showing their overdue duration and outstanding fine.	System
FR-18	Students shall be able to view notifications inside the system.	Student
Librarian operations		
FR-19	Librarians shall be able to view reservation requests and associated student identifiers.	Librarian

ID	Requirement	Role
FR-20	Librarians shall be able to mark reserved copies as ready for collection.	Librarian
FR-21	Librarians shall confirm book hand-over and record the loan start and due dates.	Librarian
FR-22	Librarians shall be able to view borrowing periods and active loans by student.	Librarian
FR-23	Librarians shall record returned copies and update their status.	Librarian
FR-24	Authorised librarians shall be able to manage book-copy inventory and record damaged or lost books.	Librarian
Student loan management		
FR-25	Students shall be able to view current and previous loans.	Student
FR-26	Students shall be able to view due dates, reservation statuses and outstanding fines.	Student
FR-27	The system shall calculate overdue penalties according to configured university policy.	System

Testable language. Compare "the system should be fast and user-friendly" (untestable) with FR-16: a tester can set a due date, advance the clock and check that exactly four reminders were sent on the right days.

4 Business rules

Proposed example policies

Business rules are the organisation's policies that the software must enforce. The values below are **illustrative** and must be confirmed by library management before implementation.

ID	Rule	Enforced by / related FR
BR-01	Loan duration defaults to **14 days**.	Configurable policy · FR-21
BR-02	Reservations expire after **48 hours** without collection (counted from the "ready" notification).	Scheduled job · FR-14
BR-03	Fine rates are configured by authorised staff (e.g. an example rate of 0.25 JOD per overdue day).	Admin policy · FR-27
BR-04	A student cannot reserve a copy already reserved by another student.	Transaction check · FR-10
BR-05	Physical collection is mandatory; no book is delivered.	Process · FR-13, FR-21
BR-06	The loan period begins when the librarian confirms hand-over, not when the student creates a reservation.	FR-21
BR-07	Returned copies must be checked against the waitlist before becoming publicly reservable.	FR-11, FR-15, FR-23
BR-08	Invalid or inactive students cannot create reservations.	FR-06
BR-09	A student cannot hold conflicting active reservations for the same title unless university policy explicitly allows it.	Transaction check
BR-10	Reservation, loan and copy states must remain synchronised.	Atomic transactions

Race conditions — why “check, then save” is not enough

Sara and Yazan both open the last available copy of *Introduction to Algorithms*. At 10:00:01 both apps check: “1 copy available”. Both press **Reserve**. If the system checks availability and saves the reservation as two separate steps, **both** requests can pass the check — and two students are promised one book.

Requirement: the availability check and the reservation write must happen inside **one atomic database transaction**, protected by a uniqueness rule (only one active reservation per copy). The first request wins; the second receives a clear message and an offer to join the waitlist. UI checks alone can never prevent this, because the conflict happens on the server.

5 Non-functional requirements

Proposed acceptance targets

Non-functional requirements describe **how well** the system must work. Targets are proposals for discussion; final values are agreed with stakeholders.

ID	Quality	Requirement	Proposed acceptance target
NFR-01	Performance	Search results return quickly under normal load.	95% of ordinary searches complete within 2 s
NFR-02	Security	All traffic encrypted; staff functions require staff roles; sessions expire safely.	HTTPS only; role checks on every protected request
NFR-03	Accessibility	Usable with keyboard, screen readers and zoom.	WCAG 2.2 Level AA target
NFR-04	Privacy	Students see only their own records; minimal personal data stored.	No other student’s data visible to students
NFR-05	Usability	A first-time student can reserve a book without help.	≥ 90% task success in usability testing
NFR-06	Responsiveness	Layout adapts to the device.	Supports mobile, tablet and desktop layouts
NFR-07	Reliability	No lost or duplicated reservations after failures.	0 duplicate active reservations per copy
NFR-08	Availability	System available during library service hours and beyond.	99.5% monthly availability
NFR-09	Scalability	Handles peaks at the start of semester and exam periods.	Target peak load agreed with IT before launch
NFR-10	Maintainability	Policies change without code changes.	Loan days, fine rates, hours are configurable
NFR-11	Auditability	Staff actions are traceable.	Every hand-over/return/fine change logged with who & when
NFR-12	Data integrity	Reservation, loan and copy states remain consistent.	Atomic database transactions prevent inconsistent reservations

6 University and external requirements

These come from outside the project team. This document **does not claim** legal compliance or certification; each item must be verified with the responsible university office.

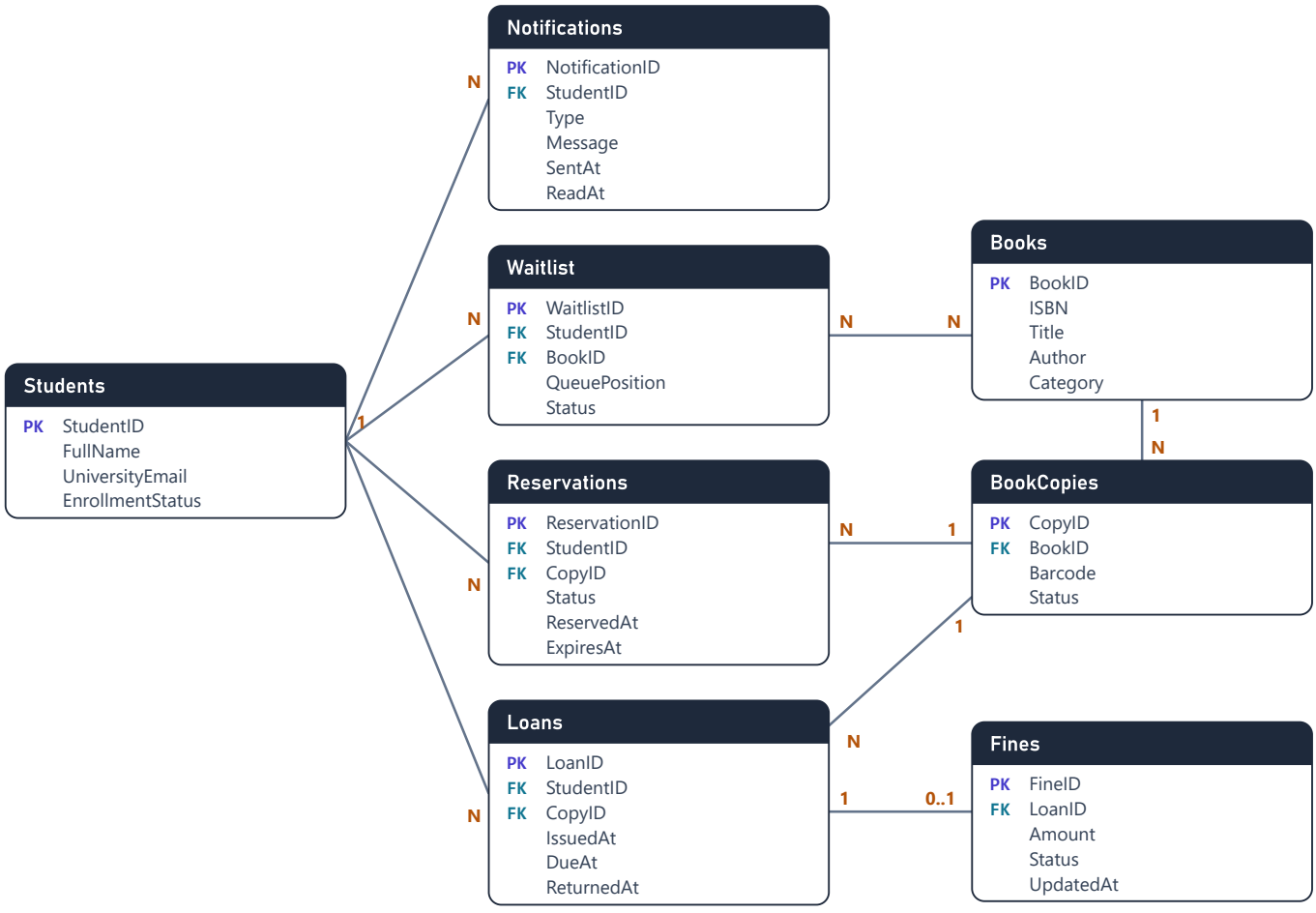
#	Area	What it means for the system	Status
1	Identity & enrolment verification	Students sign in with the university’s approved identity service; enrolment status decides eligibility (FR-05, FR-06).	Mandatory*
2	Integration with university systems	Read-only access to authorised student information systems via approved interfaces; no copying of unneeded data.	Mandatory*

#	Area	What it means for the system	Status
3	Borrowing & disciplinary policies	Loan limits, fines and blocks follow official university policy, configured by administrators (BR-01–03).	Proposed
4	Jordanian personal-data protection	Handle student data according to applicable Jordanian law (e.g. the Personal Data Protection Law, No. 24 of 2023). Scope and obligations to be confirmed by the university's legal office.	To verify
5	Accessibility standards	Design and test against WCAG 2.2 Level AA (NFR-03).	Proposed
6	Bibliographic metadata standards	Use MARC 21 or Dublin Core where catalogue data is exchanged with other library systems.	Proposed
7	Calendars, holidays & hours	Due dates and 48-hour collection windows skip configured closures; collection hours are shown to students.	Proposed
8	Retention, auditing & permissions	Define how long loan history and logs are kept; restrict access by role (NFR-11).	Mandatory*
9	Notification & communication policy	Approved channels (in-app, university e-mail), wording and frequency of reminders (FR-14–18).	Proposed

* **Mandatory** = assumed in this teaching case to have been decided by the project sponsor (library management and university IT). **Proposed** = the team's recommendation awaiting approval. **To verify** = depends on external rules that must be checked by qualified staff.

7 Database requirements (conceptual)

A conceptual model shows **what data** the system needs and how it relates — not how tables are implemented. No SQL is part of this specification.



PK primary key · **FK** foreign key · **1 / N** one-to-many · **0..1** at most one

Supporting entity	Purpose (summary)
StaffAccounts	Librarians and administrators: StaffID, name, role, active flag. Linked to the university identity service; roles drive permissions.
Policies	Configurable values with effective dates: loan days, reservation expiry hours, fine rate, maximum loans, collection hours, closures.
AuditLogs	Append-only record of important actions: who (StaffID/StudentID), what (action, entity, old → new status), when (timestamp).

Primary key (PK) — a value that uniquely identifies one row, e.g. `CopyID`.

Foreign key (FK) — a column that points to another entity's PK, e.g. `Loans.CopyID → BookCopies.CopyID`.

One-to-many — one student can have many loans; each loan belongs to exactly one student.

Books vs. copies — "Clean Code" is *one* Book with *several* BookCopies. Students reserve and borrow **copies**; they wait for **books** (any copy).

Referential integrity — an FK must always point to an existing row; a copy cannot be deleted while a loan refers to it.

Atomic transactions — related changes (reservation → loan, copy status) succeed or fail together (BR-10).

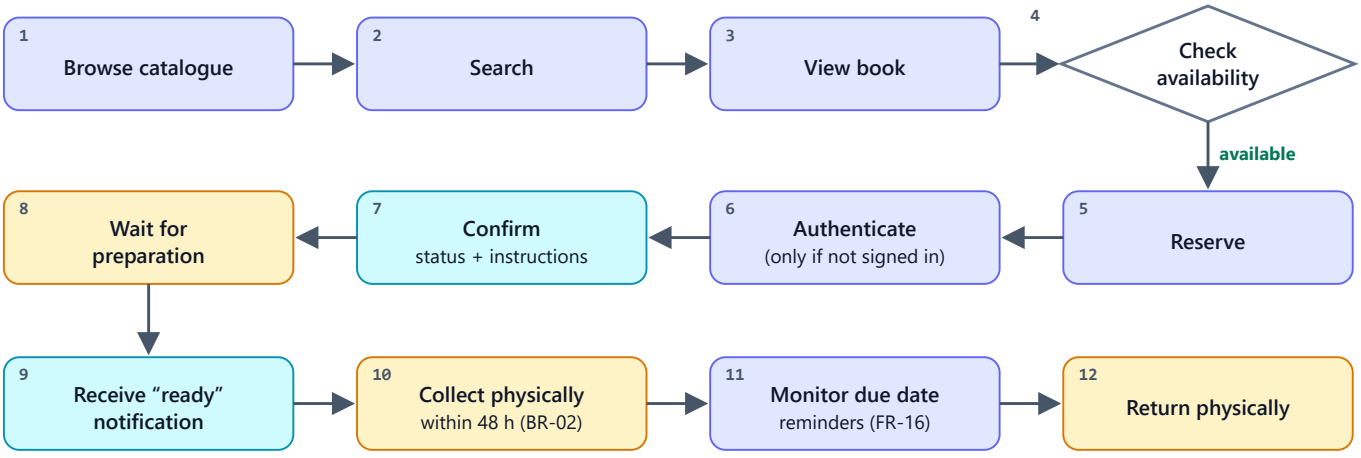
No duplicate active reservations — a uniqueness rule allows at most one *active* reservation per `CopyID` (FR-10, BR-04).

Dates & times — store timestamps in UTC with time-zone awareness; display in local time (Asia/Amman). Due dates are calculated with the academic calendar.

8 User workflows

Digital step (app) · Physical step (at the library) · System / notification · Decision

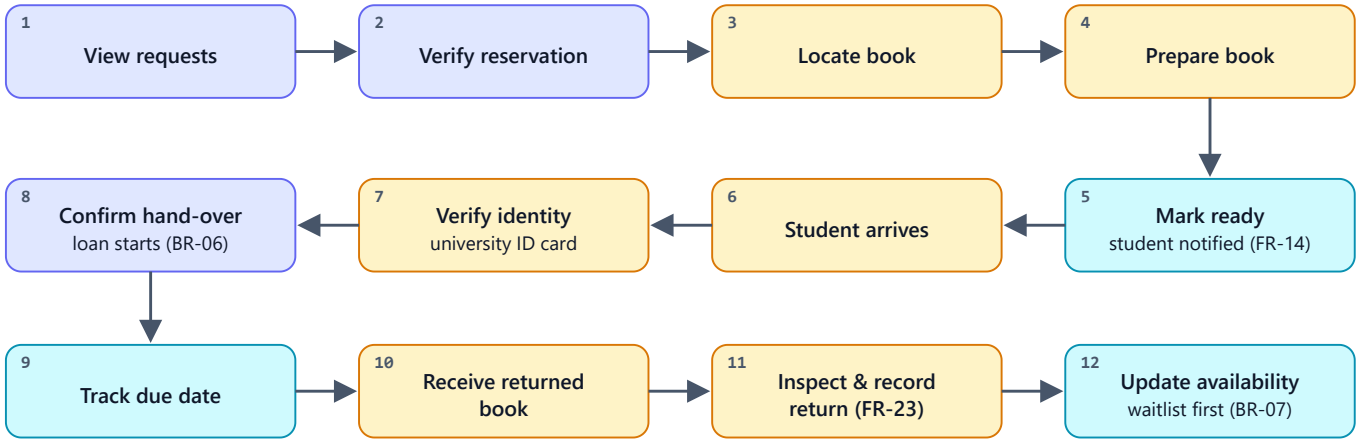
8.1 Student workflow



Alternate path — no copy available at step 4

Join waitlist (sign in if needed) → see queue position → copy returned & allocated (BR-07) → notified (FR-15) → continue at step 8

8.2 Librarian workflow



9 Edge cases and expected behaviour

#	Scenario	Expected system behaviour
E1	Two students reserve the final copy simultaneously	One atomic transaction succeeds; the other student sees “This copy was just reserved” and is offered the waitlist (FR-10, BR-04).
E2	Student authentication fails	Clear, non-technical message; retry or help link; the selected book is kept for after sign-in (FR-08).
E3	Student becomes ineligible while a reservation is pending	Reservation is flagged for the librarian and not handed over; the student is notified with the reason and next steps (BR-08).
E4	Reservation expires before collection	After 48 h the reservation becomes “Expired”, the student is informed and the copy goes to the next waitlisted student or back to available (BR-02, BR-07).
E5	A student does not receive a notification	All notifications are also visible in the in-app list (FR-18); failed deliveries are logged and retried.
E6	A book is reported lost or damaged	Librarian records the condition (FR-24); the copy leaves circulation; affected reservations are reallocated or the students notified.
E7	A student returns a book late	Return is recorded normally; the fine is calculated by configured policy (FR-27) and shown with days overdue and rate.
E8	Connectivity is lost while reserving	No reservation is created unless the server confirms it; the app shows the true status on reconnect and never shows a false success.
E9	An overdue book is returned while a fine is outstanding	The loan closes; the fine stays “Outstanding” until settled under university policy, and remains visible to the student.
E10	A copy is returned while students are waiting for that title	The copy is allocated to the first eligible waitlisted student before anyone else can reserve it (BR-07, FR-15).

From requirements to design. Every screen in the UI/UX exercise should trace back to at least one requirement above. When you critique or redesign a screen, ask: *which FR, BR or NFR does this element serve — and does the design make that requirement easy for a real user?*